

Rugged is no longer special; it's expected

SBCs no longer operate in air-conditioned control rooms. Today, they are deployed outdoors, embedded deep inside machines, mounted on vehicles, and installed near high-voltage equipment—environments where heat, vibration, EMI, and unstable power conditions are part of daily operation. As a result, ruggedisation has shifted from being a special request to a default expectation.

Support for environments ranging from -40°C to +85°C increasingly relies on industrial-grade components and fanless thermal designs. Ruggedness is no longer limited to the board itself. Increasingly, the enclosure itself is thermally engineered as a heatsink, with sealed systems expected to operate in the presence of dust, humidity, and electrical noise.

This shift is clearly reflected in current product portfolios. ADLINK's SBC-35 ASL targets fanless, low-power edge deployments where thermal headroom and long-term reliability are non-negotiable. Kontron's industrial SBC platforms, widely used in rail and automation systems, emphasise extended temperature operation, mechanical robustness, and life cycle stability. Likewise, Advantech's AIMB series is designed to withstand harsh industrial conditions while maintaining predictable performance over extended deployments.

The message for engineers is increasingly unambiguous: if an SBC cannot survive in its intended environment, raw performance numbers are irrelevant. Reliability, thermal resilience, and environmental tolerance now sit alongside compute capability as primary design criteria, especially as SBCs move closer to the physical processes they monitor and control.

SBCs are now software platforms

Hardware specifications alone are no longer the deciding factor in

SBC design wins. Today's engineers evaluate platforms through a much more demanding lens, asking questions about operating system support, BSP longevity, driver update policies, validated AI software stacks, and official middleware support, including frameworks such as ROS and ROS 2.

To address these changes, top SBC providers are now competing on software layers and support models, rather than just compute performance. For example, ADLINK supports operating systems including Linux, Ubuntu, and Yocto, along with AI development toolsets, such as OpenVINO, TensorRT, TensorFlow and Python, plus native support for ROS and ROS 2, supported through long-term support releases. Long-term support is essential in robotics, automation, and mobile applications.

At the higher end of this vendor spectrum, Nvidia's Jetson line offers comprehensive out-of-the-box software integration through AISE. This hardware-software stack is a major differentiator and one reason it is widely adopted in robotics and autonomous vehicle applications. Advantech and Kontron also compete strongly through OS stability, BSP longevity, and disciplined update schedules.

The subtext is straightforward: an SBC with unsupported software quickly degrades into technical debt. Without continuous driver updates, security patches, and validated software frameworks, even the most capable hardware becomes hard to maintain, expensive to update, and risky to deploy at scale.

Software longevity in modern SBC selection is no longer secondary but a core architectural requirement. Most modern industrial SBCs support Linux, Yocto, and Ubuntu, and often include community or vendor BSPs—but guaranteed long-term support over five to ten years should be validated on a per-vendor contractual basis. In fact, Nvidia

Jetson platforms integrate an SDK alongside JetPack support; this is strong evidence of the maturity of the software stack.

I/O, connectivity, and camera-centric design are scaling rapidly

Modern SBCs are increasingly evolving into data aggregation and sensor-fusion platforms, rather than pure compute boards typical of earlier SBC generations. This transition is evident across both ARM- and x86-based ecosystems, where I/O architecture has become a primary differentiator, often outweighing raw CPU performance in real-world deployments.

High-speed expansion is now a baseline expectation. Platforms such as Nvidia Jetson Orin NX and Orin Nano, Advantech's SOM-7583, and ADLINK's AmITX-RK3588 already support PCIe Gen 4, with vendor roadmaps extending towards PCIe Gen 5 in upcoming industrial SBC and COM designs. This bandwidth is essential for attaching AI accelerators, NVMe storage, and multi-camera capture cards without creating bottlenecks in edge inference pipelines.

Camera connectivity, in particular, is scaling rapidly. Boards such as Jetson AGX Orin, Toradex Verdin i.MX8M+, AAEON UP Squared Pro 7000, and PHYTEC phyBOARD-Pollux now expose native MIPI CSI lanes for direct connections to camera modules with high-resolution image sensors. Rather than utilising USB camera connections, native MIPI reduces latency and CPU load, which is increasingly critical in robotics, vision inspection systems, and automotive applications.

In the SBC space, deeper industrial penetration continues alongside increasing Ethernet density, including dual- and quad-Ethernet configurations, 2.5 GbE, and time-sensitive networking support on SBCs from companies such as IEI Integration, AAEON, and Kontron.